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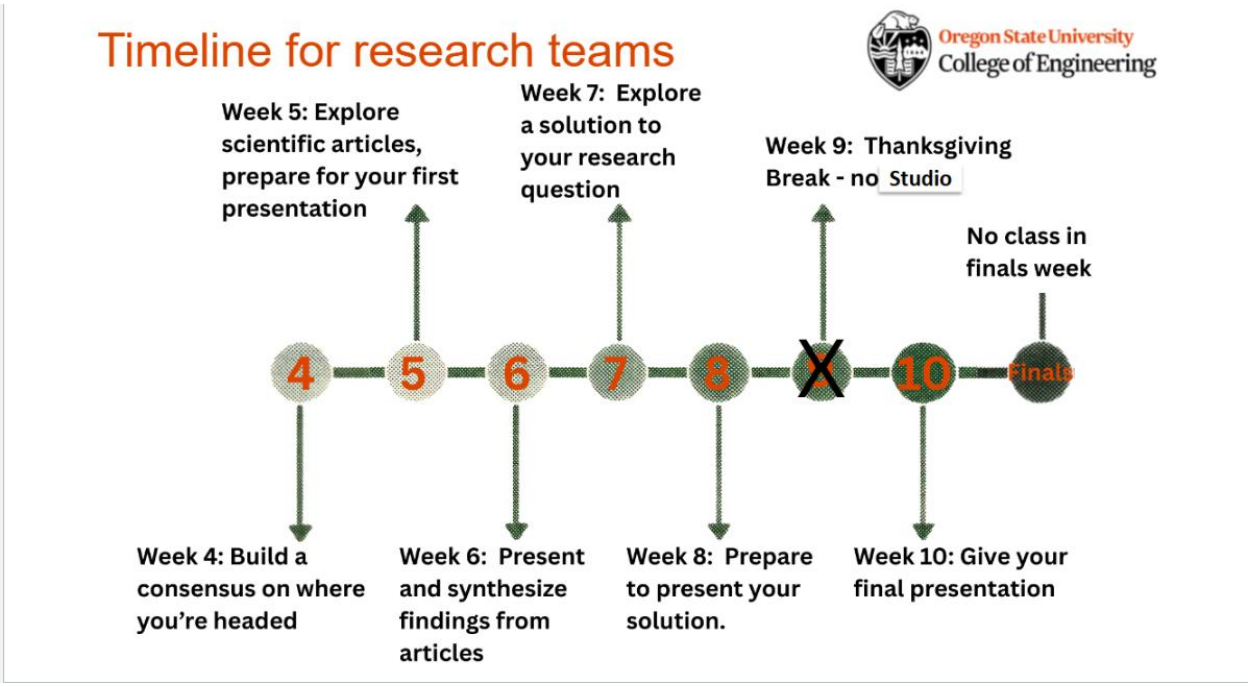
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Adaptive Control under Input/State Constraints: *How can we guarantee Bounded-Input Bounded-State (BIBS) stability in highly non-linear electric vehicle powertrains when subject to sudden, unmodeled actuator saturation?*

Topics covered: Data and signal transmissions in electrical engineering, chaos theory in mathematics, simulations/models in computer science



Week 6 Presentation requirements

- Topic Introduction
- Problem Statement

- Approach
- Proposal Objective
- How will we collaborate
- Conclusion
- References(APA Citations)

Coding – Rudolph, Liam

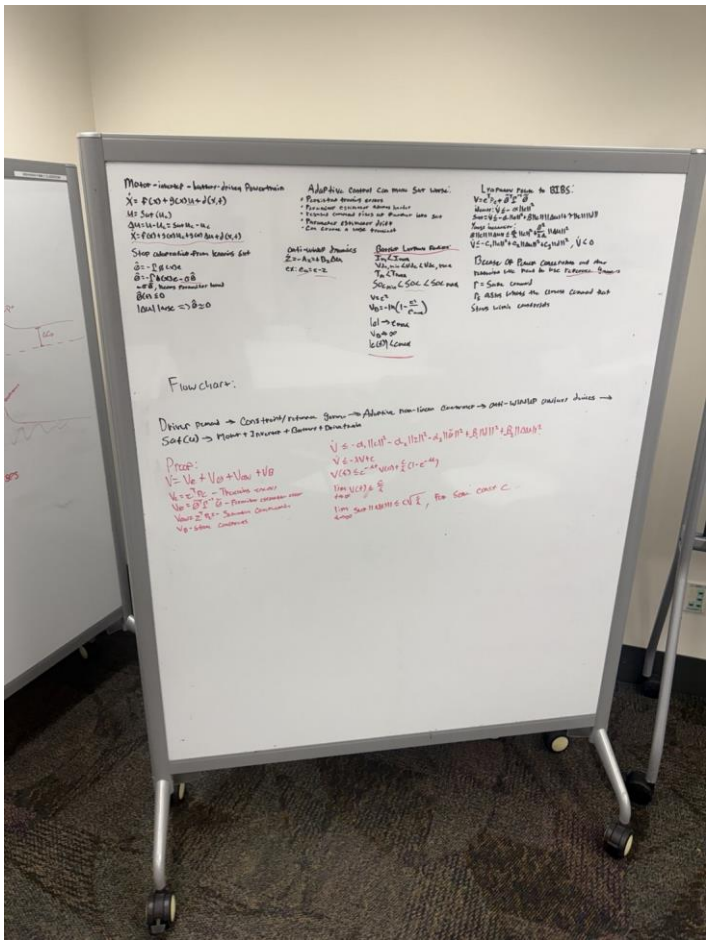
Theoretical – Nolan, Kai

Data Analysis: Saleheen, Taehwan

Adaptive Control under Input/State Constraints

Working notes for the EV powertrain / BIBS project

Goal: get enough control theory in place to actually model, simulate, and prove something useful - not learn an entire controls curriculum first.



Whiteboard from our project discussion. This guide follows the same flow, just cleaned up.

The whole project in one sentence: If the adaptive controller asks for an input the real EV hardware cannot deliver, we want the system to stay bounded, keep important states inside safe limits, and recover cleanly instead of letting the adaptation run away.

What I think we actually need to know

- State-space models: x-dot = f(x,u), what the states mean physically, and how feedback changes the dynamics.
- Lyapunov stability: how to turn "this should stay stable" into an inequality we can actually prove.
- Adaptive control: how parameter estimates change online, and why saturation can fool the estimator.
- Anti-windup / saturation compensation: how to separate actuator failure-to-follow from real model error.
- State constraints: how to keep current, voltage, SOC, temperature, etc. inside limits.
- Reference governors: how to stop asking the inner controller to track something physically impossible.

Start with the plant model

For now I would keep the EV model generic and only add complexity when we need it: $\dot{x} = f(x) + g(x)u + d(x,t)$

Possible states later: motor speed, q/d-axis currents, DC-link voltage, shaft twist, battery SOC, battery or motor temperature. The exact list depends on how detailed we decide to make the first model.

Where saturation enters

The controller computes a command u_c, but the actuator applies something else once a limit is hit: $u = \mathrm{sat}(u_c)$ and $\Delta u = u - u_c$

That Δu term is the key. If the proof silently assumes $u = u_c$, the proof can stop being valid as soon as the motor/inverter saturates.

EV example: Controller asks for 450 N·m, but the motor/inverter can only give 300 N·m. Tracking error stays nonzero even if the controller model is otherwise correct.

BIBS / UUB - what “stable enough” means here

$$||u(t)|| \le \bar{u} \quad \Rightarrow \quad ||x(t)|| \le x$$

BIBS means a bounded input cannot drive the state to infinity. For this project, uniform ultimate boundedness (UUB) may be the more realistic theorem because persistent disturbances and saturation can prevent perfect tracking.

$$\text{after the transient:} \quad ||x(t)|| \le \text{some calculable bound}$$

So the goal is not necessarily “error goes exactly to zero no matter what.” The useful claim is closer to: saturation can hurt performance, but it cannot make the closed-loop states blow up under our stated assumptions.

Lyapunov theory - the main proof tool

Pick an energy-like function V that is positive away from the desired state. Then show its time derivative is negative except for bounded disturbance terms.

$$V > 0, \quad V(0)=0, \quad \dot{V} < 0$$

For adaptive control, V normally includes both tracking error and parameter-estimation error:

$$V = e^{T}Pe + \tilde{\theta}^{T} \Gamma^{-1} \tilde{\theta}$$

The inequality I want us to keep coming back to

The clean end goal is something like:

$$\dot{V} \le -\lambda V + c$$

This solves to

$$V(t) \le e^{-\lambda t} V(0) + (c/\lambda)(1 - e^{-\lambda t})$$

and therefore $\limsup V(t) \le c/\lambda$

Why this matters: If V bounds the physical states, then c/λ gives us a route to an actual state bound. This is the bridge from “bounded saturation/disturbance” to BIBS/UUB.

Why naive adaptive control can go wrong

A basic adaptive law might look like $\dot{\hat{\theta}} = -\Gamma$

$$\phi(x)e$$

During saturation, the error can stay large because the actuator physically cannot follow u_c. The adaptive law may mistake that error for a bad parameter estimate and keep changing $\hat{\theta}$. That is the adaptive version of windup.

Ways to stop the estimator from chasing saturation

- σ-modification: add leakage so estimates do not drift indefinitely.

$$\dot{\hat{\theta}} = -\Gamma \phi(x) e - \sigma \hat{\theta}$$

- Projection: force θ-hat to stay inside a physically reasonable parameter set.
- Adaptation gating / dead zone: slow or freeze learning when the error is obviously being driven by saturation or noise.

Put saturation directly into the proof

$$\dot{x} = f(x) + g(x)u_c + g(x)\Delta u + d$$

Now Δu is explicit instead of hidden. A Lyapunov derivative might end up with a cross term like

$$\dot{V} \le -\alpha ||e||^2 + \beta ||e|| ||\Delta u|| + \gamma ||e|| ||d||$$

Use Young's inequality to absorb the cross terms:

$$\beta ||e|| ||\Delta u|| \le (\alpha/2)||e||^2 + (\beta^2/(2\alpha))||\Delta u||^2$$

$$\dot{V} \le -c_1 ||e||^2 + c_2 ||\Delta u||^2 + c_3 ||d||^2$$

Anti-windup: give the controller a way to account for Δu

One simple idea is an auxiliary state z driven by the saturation mismatch: $\dot{z} = -A_z z + B_z \Delta u$

Then use a compensated error such as $e_a = e - Cz$ in the controller or adaptive law. The idea is not that z magically fixes saturation; it tells the estimator "this part of the error came from the actuator limit."

What I would compare in simulation: Same plant and same saturation event, once with naive adaptation and once with robust adaptation + anti-windup. Plot tracking error, parameter estimates, u_c, u, and Δu.

State constraints: BIBS alone is not enough

An EV can be bounded and still violate something important. We may also need guarantees like:

$$I_m \le I_{max} \qquad V_{dc,min} \le V_{dc} \le V_{dc,max} \qquad SOC_{min} \le SOC \le SOC_{max}$$

Barrier Lyapunov functions are useful because they become very large as a state approaches a forbidden boundary. A common scalar example is

$$V_B = -\ln(1 - e^2/e_{max}^2)$$

As |e| approaches e_max, V_B goes to infinity. If our proof keeps V_B finite, then the boundary cannot be crossed, assuming the initial condition starts inside the feasible set and a feasible input exists.

Reference governor: don't ask the inner loop to do the impossible

driver request r → reference governor → safe request r_safe

If the requested torque/braking would violate current, voltage, battery power, SOC, or thermal limits, the governor modifies the request before it reaches the adaptive controller. That makes the inner-loop problem much cleaner.

Important limitation: No controller can guarantee perfect tracking and every constraint for an arbitrary impossible command. We need a feasible control set. The reference governor is one way to handle that at the command level.

The architecture I think we should try first

Driver demand → Constraint / Reference Governor → Adaptive nonlinear controller

→ Anti-windup compensation → Saturated actuator → EV

powertrain Then build the proof around a composite Lyapunov function:

$$V = V_e + V_\theta + V_{aw} + V_B$$

$V_e = e^T P e$ $V_\theta = \tilde{\theta}^T \Gamma^{-1} \tilde{\theta}$ V_aw =

$z^T P_z z$ The proof target is something in this family:

$$\dot{V} \le -\alpha_1 ||e||^2 - \alpha_2 ||z||^2 - \alpha_3 ||\tilde{\theta}||^2 + \beta_1 ||d||^2 + \beta_2 ||\Delta u||^2$$

$$\Rightarrow \text{after bounding terms: } \dot{V} \le -\lambda V + c$$

First simulation path

Step	Build	What we learn
1	Second-order nonlinear toy plant	One unknown parameter + one hard actuator limit. Get the math working before touching a full EV model.
2	Naive adaptive controller	Show parameter drift / bad recovery when saturation occurs.

3	Robust adaptation	Add σ-modification or projection and compare parameter behavior.
4	Anti-windup state	Add z dynamics and compare overshoot + recovery after saturation clears.
5	One state constraint	Add a BLF and verify the constrained state stays inside its bound.
6	Simplified EV model	Move to longitudinal vehicle + motor/inverter dynamics; add battery/thermal states only if needed.

Plots I would always save: u_c vs. u, Δu, tracking error, important states, parameter estimates, V(t), estimated V-dot, and distance to each active constraint.

Two-week ramp-up plan

When	Topic	Checkpoint
Days 1-2	State-space + Lyapunov basics	Be able to explain x-dot=f(x,u), equilibrium, positive-definite V, and why V-dot<0 matters.
Days 3-4	BIBS / ISS / UUB	Understand what kind of boundedness theorem we actually want and what assumptions it needs.
Days 5-6	Adaptive control	Work through one scalar/secondorder adaptive example and derive θ-hat-dot from a Lyapunov function.
Days 7-8	Saturation + robustness	Add saturation. See parameter drift. Add σ-modification / projection.
Days 9-10	Anti-windup	Add Δu-driven auxiliary dynamics and compare recovery.
Days 11-12	Barrier functions	Work one constrained scalar example before putting a BLF into the EV model.

Days 13-14	Reference governor + project model	Decide what our first EV state vector, actuator limit, and research theorem will actually be.

References - I would read these in this order

Khalil, Nonlinear Systems, 3rd ed. Use it for Lyapunov stability, nonlinear systems, and input-to-state ideas. Do not read cover-to-cover.

Ioannou & Sun, Robust Adaptive Control. Focus on robustness modifications: σ-modification, projection, dead zones, and unmodeled dynamics.

Tee, Ge & Tay, Automatica (2009), "Barrier Lyapunov Functions for the control of output-constrained nonlinear systems." Good first BLF paper. DOI 10.1016/j.automatica.2008.11.017.

Gilbert, Kolmanovsky & Tan (1995), discrete-time reference governors. Foundation for modifying references to respect state/control constraints. DOI 10.1002/rnc.4590050508.

Garone, Di Cairano & Kolmanovsky, Automatica (2017), reference/command governor survey. Read after the basic reference-governor paper for the bigger picture. DOI 10.1016/j.automatica.2016.08.013.

Questions we need to answer as a group

- What is the smallest EV model that is still interesting enough to publish/use?
- Which saturation are we actually studying first: torque, current, voltage, power, rate limit, or thermal derating?
- Is our main theorem BIBS/UUB, ISS, forward invariance of a safe set, or some combination?
- Does the admissible input set change with speed, SOC, temperature, or DC-link voltage?
- Which uncertainties are learnable parameters and which are just bounded disturbances?
- What baseline are we comparing against?
- Where is the novelty: adaptive law, saturation estimator, time-varying constraints, proof, or EV application?

Quick whiteboard cheat sheet

Plant	x-dot = f(x) + g(x)u + d
Saturation	u = sat(u_c), $\Delta u = u - u_c$

Adaptive law idea	θ-hat-dot = -Γφe (then add robustness)
Anti-windup	z-dot = -A_z z + B_zΔu
Barrier idea	V_B →∞ at the state boundary
Proof target	V-dot ≤ -λV + c
Result	bounded disturbance/saturation ⇒ bounded ultimate state region

My current takeaway: Treat saturation as part of the dynamics, do not let the adaptive law learn an impossible command, keep the important states inside a feasible set, and make the boundedness claim come directly from the Lyapunov derivative.

Team 1 — Plant & EV Modeling

This group decides **what system we are actually studying**.

Initially, do **not** build a giant EV model. Your notes specifically recommend starting with a second-order nonlinear toy plant and only later moving into a simplified EV model. Pasted text(20261026-194242)

Their first job should be something like

\[\dot{x}=f(x)+g(x)u+d(x,t) \]

with one unknown parameter and one actuator saturation.

Eventually, they decide which actual EV states matter. Candidates from your notes are motor speed, \(q/d\)-axis currents, DC-link voltage, shaft twist, battery SOC, and thermal states.

Their immediate questions are:

- Are we studying **torque saturation**, current saturation, voltage saturation, power limits, or thermal derating?
- What parameters are uncertain?
- What disturbances are external?
- What actuator limits are physically reasonable?
- What is the smallest EV model worth publishing?

I'd probably start with **torque saturation** because it's easiest to understand:

$$ u_c = 450\text{ N·m} $$

but

$$ u = 300\text{ N·m} $$

so

$$ \Delta u=u-u_c=-150\text{ N·m}. $$

That saturation mismatch Δu becomes extremely important later.

Team 2 — Adaptive Control & Anti-Windup

This group creates the thing controlling Team 1's model.

Start with a deliberately **naive adaptive controller**.

Something like

$$ \dot{\hat{\theta}}=-\Gamma\phi(x)e. $$

Then intentionally saturate the actuator and watch the parameter estimator start behaving badly.

That's important because you want to demonstrate the problem before presenting the solution. Your notes specifically identify the issue: saturation can keep tracking error high, causing the adaptive law to mistakenly interpret actuator limitations as parameter error. Pasted text(20261026-194242)

Then this team tests:

σ-modification

$$ \dot{\hat{\theta}} = -\Gamma\phi e-\sigma\hat{\theta} $$

Projection

Keep the estimated parameter inside a known physical range.

Adaptation gating

Freeze or slow adaptation while severe saturation is occurring.

Then build the anti-windup system:

$$ \dot z=-A_z z+B_z\Delta u $$

and possibly use

$$ e_a=e-Cz. $$

The basic concept is:

"Hey adaptive controller, some of this tracking error is because the actuator hit its limit. Don't blame the model."

That architecture comes directly from the project notes. Pasted text(20261026-194242)

Team 3 — Stability & Constraints

This is probably the most mathematically challenging team.

Their job is answering:

Can we prove that the system doesn't go to infinity?

Start with a composite Lyapunov function such as

$$ V=e^TPe+\tilde\theta^T\Gamma^{-1}\tilde\theta. $$

Eventually add the anti-windup state and barrier function:

$$ V=V_e+V_\theta+V_{aw}+V_B. $$

The target theorem should look something like

$$ \dot V\leq-\lambda V+c. $$

Then

$$ \limsup_{t\to\infty} V(t)\leq\frac{c}{\lambda}. $$

That gives you an actual calculable ultimate bound instead of just saying "our simulations look stable." Pasted text(20261026-194242)

I'd have this team study **UUB first**, not obsess over asymptotic stability.

Because saturation and disturbances may prevent

$$ e(t)\rightarrow0. $$

But you may still be able to prove

$$ \|x(t)\|\leq B $$

after the transient.

That's potentially the central theoretical result of the whole project. Pasted text(20261026-194242)

This group also handles state constraints with Barrier Lyapunov Functions.

For example,

$$ |I_m|<I_{\max} $$

or

$$ SOC_{\min}<SOC<SOC_{\max}. $$

BIBS alone isn't enough because an EV could technically remain "bounded" while still overheating a motor or exceeding safe current. Pasted text(20261026-194242)

Team 4 — Simulation & Validation

This team becomes the glue holding everyone together.

They create one common simulation environment so everyone isn't writing completely separate models.

I would make their experiment matrix:

Experiment A — No saturation

Verify the baseline controller works.

Experiment B — Saturation + naive adaptive controller

Show parameter drift and poor recovery.

Experiment C — Robust adaptive controller

Add σ-modification or projection.

Experiment D — Anti-windup

Add the Δu-driven compensation.

Experiment E — State constraint

Introduce a barrier function.

Experiment F — EV model

Move the successful architecture onto the simplified EV system.

That's essentially the progression laid out in the working notes. Pasted text(20261026-194242)

For every simulation, automatically save:

$$ u_c,\quad u,\quad \Delta u $$

tracking error,

state trajectories,

parameter estimates,

$$ V(t), $$

estimated

$$ \dot V, $$

and distance from each constraint.

Those plots will eventually become paper figures.

What the whole team actually needs to accomplish

I would turn the project into **six major milestones** rather than saying "research adaptive control."

Milestone 1 — Define the exact research problem

You guys need one sentence such as:

Can an adaptive controller with saturation-aware anti-windup and robust parameter adaptation guarantee UUB stability of a nonlinear EV drivetrain under actuator torque saturation?

Then decide:

$$ \boxed{\text{Plant + uncertainty + saturation + theorem}} $$

Right now this is the most important task.

Your own notes still list these as unanswered questions: which saturation to study, whether the main result should be BIBS/UUB/ISS/forward invariance, which uncertainties are learnable, what baseline to compare against, and where the novelty actually sits. Pasted text(20261026-194242)

Milestone 2 — Build the toy problem

Before touching a realistic EV:

$$\boxed{\text{second-order nonlinear plant}}$$

plus

$$\boxed{\text{one unknown parameter}}$$

plus

$$\boxed{\text{one actuator limit}}$$

That becomes the sandbox where everybody learns the theory.

Milestone 3 — Reproduce the failure

This is important for a paper.

Show:

normal system → works

then

saturation → adaptive estimator goes crazy

You want a figure that makes someone immediately understand why the research problem matters.

Milestone 4 — Build your proposed controller

Something like:

Driver Demand

↓

Constraint / Reference Governor

↓

Adaptive Nonlinear Controller

↓

Anti-Windup Compensation

↓

Actuator Saturation

↓

EV Plant

Milestone 5 — Prove the result

Get everything into something resembling

\[\dot V \le -\alpha_1\|e\|^2 -\alpha_2\|z\|^2 -\alpha_3\|\tilde{\theta}\|^2 +\beta_1\|d\|^2 +\beta_2\|\Delta u\|^2. \]

Then reduce it to

dot V lambda V+c

and derive the ultimate state bound.

That is where this transitions from **"cool controller simulation"** into legitimate control-theory research.

Milestone 6 — Put it on an EV model

Only after 1–5 work.

Then replace the toy system with something like:

vehicle longitudinal dynamics + motor dynamics + inverter/torque limit

Battery and thermal states can come afterward if they're actually necessary.

How I would divide the first week

If you guys meet this week, I'd literally write these four boxes on the board:

Model Team:
"Produce our first nonlinear plant and actuator saturation equation."

Controls Team:
"Produce baseline adaptive controller + σ-modification."

Theory Team:
"Derive candidate and start deriving."

Simulation Team:
"Build the shared simulation framework and plotting."